MACHINE TOOLS

III B. Tech. II Semester: MECHANICAL ENGINEERING								
Course Code	**Category**	**Hours / Week**			**Credits**	**Maximum Marks**		
A6ME33	**PCC**	**L**	**T**	**P**	**C**	**CIE**	**SEE**	**Total**
		3	0	0	3	40	60	100

COURSE OVERVIEW:

This course will give an insight over different machining processes and tools. In the metal-working industry work pieces of most different shapes and dimensions and of different materials are worked. The shaping of materials is done by either non-cutting process or cutting process. Such relative motion is produced by combination of rotary and translating movements either of work piece or the cutting tool or of both. Machining process also include other processes like grinding, slotting, shaping, honing, planning, lapping, and broaching operations.

COURSE OUTCOMES:

At the end of the course, the student should be able to:

1. Knowledge on cutting tool geometry, mechanism of chip formation and mechanics of orthogonal cutting.
2. Apply the fundamentals and principles of metal cutting to practical applications using lathes.
3. Demonstrate the fundamentals of machining processes and machine tools of Shaping, Slotting, Plaining and Drilling machines.
4. Demonstrate the fundamentals of machining processes and gear cutting using Milling Machine.
5. Select appropriate super finishing process for an application and understand the impact of such process in environmental context.

UNIT-I **METAL CUTTING THEORY**

Elementary treatment of metal cutting theory – Elements of cutting process – Geometry of single point cutting tool and angles. Chip formation and types of chips – built up edge and its effects. Mechanics of orthogonal cutting, – Merchant’s Force diagram – cutting speeds, feed, depth of cut, tool life, coolants, machinability – Tool materials.

UNIT-II **LATHE**

Engine lathe – Principle of working, specification of lathe – types of lathes – work & tool holding devices-Taper turning, thread- turning – for Lathes and attachments. Turret and capstan lathes – Principal features of automatic lathes – classification – Single spindle and multi-spindle automatic lathes – tool layout

UNIT-III **SHAPING SLOTTING AND PLANING**

Shaping slotting and planing machines – Principles of working – Principal parts – specification classification, operations performed.

Drilling and Boring Machines – Principles of working, specifications, types, operations performed – tool holding devices – twist drill – Boring machines – Fine boring machines – Jig Boring machine. Deep hole drilling machine.

UNIT-IV **MILLING MACHINE**

Milling machine – Principles of working – specifications – classifications of milling machines – Principal features of horizontal, vertical and universal milling machines – machining operations , Geometry of milling cutters – Types of milling cutters – Accessories to milling machines.

UNIT-V	SUPER FINISHING PROCESS
Grinding machine – Fundamentals – Theory of grinding – classification of grinding machine – cylindrical and surface grinding machine – Tool and cutter grinding machine – special types of grinding machines – Different types of abrasives – bonds specification of a grinding wheel and selection of a grinding wheel. **Lapping, honing and broaching machines** – comparison to grinding – lapping and honing.	

Text Books:

1. R.K. Jain and S.C. Gupta, Production Technology, S Chand Publication, 2009.
2. O P Khanna, Production Technology, Danpat Rai Publications, 2015.

Reference Books:

1. G. Boothroyd G. Boothroyd, Fundamentals of Metal Machining and Machine Tools, Taylor and Francis, 2005.
2. C. Elanchezhian and M. Vijayan, Machine Tools, Anuradha Agencies, 2009.
3. Bhattacharya A and Sen.G.C., Principles of machine Tools, New central Book Agency, 2007.